2 Marks

1. **Define a file system.** Answer: A file system is a software component that manages the storage, organization, retrieval, naming, and access of files on secondary storage devices. It provides a logical view of files to users and applications while mapping them to physical blocks on disk.

2. **What are the main objectives of a file system?** Answer: The objectives are: (i) to provide efficient and convenient access to files, (ii) to ensure data integrity and protection, (iii) to support multiple users and processes simultaneously, and (iv) to optimize storage utilization.

3. **Define access methods in file systems.** Answer: Access methods specify how data in a file can be read or written. The common methods are sequential access, direct (random) access, and indexed access.

4. **Differentiate between sequential and direct access methods.** Answer: Sequential access reads/writes records one after another (like magnetic tape). Direct access allows a program to read/write any record directly by giving its relative block number (like disk).

5. **What is a directory structure?** Answer: A directory structure is a logical organization of directories (folders) and files that helps users locate and manage files efficiently. It defines the hierarchy and naming scheme of files in the system.

6. **State any two advantages of tree-structured directory.** Answer: (i) Allows users to create their own sub-directories, (ii) Provides efficient searching and grouping of related files, and (iii) Supports file sharing and protection at different levels.

7. **Define file protection.** Answer: File protection is the mechanism used by the operating system to control which users or processes can perform operations (read, write, execute, append, delete) on a file or directory.

8. **What is Discretionary Access Control (DAC)?** Answer: DAC is a protection model in which the owner of a file decides who can access it and what operations they can perform. Access is granted at the owner’s discretion (e.g., UNIX chmod permissions).

9. **What is Mandatory Access Control (MAC)?** Answer: MAC is a system-enforced access control model where the operating system (not the owner) decides access rights based on security labels (classification levels) and clearances. It is used in high-security environments.

10. **What is a File Control Block (FCB)?** Answer: FCB is a data structure maintained by the operating system for every open file. It contains all the information needed to manage the file (e.g., file pointer, file permissions, location of data blocks).

11. **What is an inode in the context of file systems?** Answer: Inode (index node) is a data structure in UNIX/Linux that stores all metadata of a file (size, permissions, timestamps, block addresses) except the file name and actual data.

12. **Name the two common directory implementation techniques.** Answer: (i) Linear list (array of directory entries) and (ii) Hash table (directory entries stored in a hash table for faster lookup).

13. **List any four file allocation methods.** Answer: Contiguous allocation, Linked allocation, Indexed allocation, and FAT (File Allocation Table) based allocation.

14. **Define contiguous allocation.** Answer: In contiguous allocation, each file is stored in a set of consecutive disk blocks. It is simple but suffers from external fragmentation.

15. **What is indexed allocation?** Answer: In indexed allocation, an index block (inode or index table) contains pointers to all the data blocks of a file. It supports random access and avoids external fragmentation.

16. **What is free-space management?** Answer: Free-space management is the technique used by the file system to keep track of unused disk blocks so that new files can be allocated space efficiently.

17. **Name any two free-space management techniques.** Answer: (i) Bit vector (bitmap) and (ii) Linked list (free-list).

18. **What is a Virtual File System (VFS)?** Answer: VFS is an abstract layer provided by the operating system that allows different file systems (ext4, NTFS, FAT, etc.) to coexist and be accessed using the same system-call interface.

19. **State any two features of the Linux ext4 file system.** Answer: (i) Supports extents (contiguous blocks) for better performance, (ii) Uses journaling for crash recovery, and (iii) Maximum file size up to 16 TB and volume size up to 1 EB.

20. **What is the role of VFS in Linux?** Answer: VFS provides a common interface between user programs and various concrete file systems in Linux. It translates generic system calls into file-system-specific operations.

<u>**10 Marks**</u>

1. **Explain the various file access methods with suitable examples.**
   Answer:
   File access methods define the way records or data blocks in a file can be accessed by the user or application. The operating system provides these methods:

- Sequential Access Method: Data is accessed in a fixed sequential order, one record after another. The file pointer moves forward only. Operations: Read next, Write next. Example: Magnetic tapes, audio/video streaming files, batch processing files. Advantages: Simple to implement, fast for sequential processing. Disadvantages: Cannot jump to a particular record easily. Used in: Payroll processing, transaction logs.
- Direct (Random) Access Method: Any record/block can be read or written directly by specifying its relative block number or address. Operations: Read n, Write n, Seek to n. Example: Hard disks, database files, airline reservation systems. Advantages: Faster access to any record, supports random I/O. Disadvantages: More complex implementation. Most modern disks support this method.
- Indexed Access Method: An index is maintained that contains pointers (or keys) to the actual data blocks/records. The index is searched first to locate the required record. Example: Indexed Sequential Access Method (ISAM) used in older mainframe systems, B+ tree indexed files. Advantages: Combines benefits of sequential and direct access. Disadvantages: Extra space required for index, overhead in maintaining index.

Modern operating systems (like Linux) usually implement sequential and direct access at the kernel level, while indexed access is handled by higher-level database management systems.

2. Discuss the different directory structures used in file systems.

Answer:

Directory structures organize files and sub-directories logically.

- Single-level Directory: All files are placed in one directory. Advantages: Simple. Disadvantages: Naming conflict, difficult to manage when number of files increases.
- Two-level Directory: One directory per user (Master File Directory + User File Directories). Advantages: Solves naming conflict between users. Disadvantages: No grouping of related files of same user.
- Tree-Structured Directory (Most common): Hierarchical structure with root directory and sub-directories. Example: UNIX/Linux, Windows. Advantages: Efficient grouping, easy searching, protection at each level, supports path names (absolute & relative). Disadvantages: Cannot share files easily.
- Acyclic-Graph Directory: Allows directories to have shared sub-directories through links (hard/soft links). Advantages: File sharing is possible. Disadvantages: Requires link count or garbage collection for deletion.
- General Graph Directory: Allows cycles (a directory can point back to ancestor). Advantages: Maximum flexibility. Disadvantages: Very complex traversal and deletion. Rarely used.

Linux uses tree-structured directory with symbolic and hard links (acyclic-graph support).

3. Explain file protection mechanisms in operating systems. Differentiate between DAC and MAC with examples.

Answer:

File protection prevents unauthorized access, modification, or deletion of files.

Protection Mechanisms:

- Access Control Lists (ACLs)
- Capability Lists
- Permission Bits (UNIX style)
- Passwords for files (rare)

Discretionary Access Control (DAC): The owner of the file decides who can access it and what rights they have. Example: In Linux, chmod 755 file.txt → owner (rwx), group (r-x), others (r-x). Advantages: Flexible, easy to use. Disadvantages: Owner can accidentally give excessive rights; Trojan horse attacks possible. Used in: General-purpose OS like Windows, Linux.

Mandatory Access Control (MAC): The operating system enforces access rules based on security labels (e.g., Top Secret, Secret, Confidential) and user clearances. Owner cannot override the rules. Example: SELinux in Linux, military and government systems. A user with "Secret" clearance cannot read "Top Secret" files even if the file owner allows it. Advantages: Highly secure, prevents information leakage. Disadvantages: Rigid, difficult to administer.

Comparison: DAC is user-controlled, MAC is system-controlled. Most systems use DAC; high-security systems combine both (Linux with SELinux).

4. Describe the file system structure (layers) in detail.

Answer:

A file system is organized in layers for modularity and abstraction:

1. Application Programs Layer: Users and applications issue system calls (open, read, write, etc.).
2. Logical File System Layer: Manages directory structure, file metadata, permissions, and symbolic links. Maintains File Control Blocks (FCBs) or inodes.
3. File-Organization Module: Knows about allocation methods (contiguous, indexed), free-space management, and translates logical block numbers to physical blocks.
4. Basic File System / Virtual File System (VFS): Issues generic commands to device drivers. In Linux, VFS is present here.
5. I/O Control Layer + Device Drivers: Interfaces with hardware.

Linux VFS adds an extra abstraction layer so that multiple file systems (ext4, XFS, NFS, FAT32, etc.) can be accessed uniformly using the same system calls.

5. Compare the different directory implementation techniques. (10) Answer:

Technique	Search Time	Insertion/Deletion	Advantages	Disadvantages	Used In
Linear List	O(n)	O(1)	Simple implementation	Slow for large directories	Small file systems

Technique	Search Time	Insertion/Deletion	Advantages	Disadvantages	Used In
Hash Table	O(1) average	O(1)	Fast lookup	Hash collisions, fixed size	Many modern systems
B-Tree / B+ Tree	O(log n)	O(log n)	Balanced, efficient for large dirs	Complex implementation	ext4, NTFS, XFS

Linux ext4 uses hashed b-tree (HTree) for directory indexing, which gives fast lookup even for directories containing thousands of files.

6. **Explain the various file allocation methods with their advantages and disadvantages.**
   Answer:

- Contiguous Allocation: File occupies consecutive blocks. Advantages: Fast sequential & random access, simple. Disadvantages: External fragmentation, difficult to grow file. Example: CD-ROMs, early systems.
- Linked Allocation: Each block contains pointer to next block. Advantages: No external fragmentation, easy file growth. Disadvantages: Slow random access, pointer overhead, reliability issue (if one pointer lost, whole file lost). Example: FAT file system (partial).
- Indexed Allocation: One index block contains addresses of all data blocks. Advantages: Supports random access, no external fragmentation. Disadvantages: Index block overhead (solved by multi-level indexing). Example: UNIX/Linux inodes.
- Extent-based Allocation (Modern improvement): Instead of single blocks, contiguous chunks (extents) are allocated. Linux ext4 heavily uses extents for better performance and reduced metadata.

7. **Discuss free-space management techniques with examples.**
   Answer:

- Bit Vector / Bitmap: One bit per disk block (0 = free, 1 = allocated). Advantages: Simple, efficient for finding contiguous blocks. Disadvantages: Large bitmap size for big disks. Used in: Linux ext4.
- Linked List: All free blocks are linked together. Advantages: Easy to implement. Disadvantages: Slow to find contiguous space.

- Grouping: First free block points to a group of free blocks. Advantages: Faster than simple linked list.
- Counting: Stores starting block address + count of contiguous free blocks. Advantages: Very efficient when many contiguous free blocks exist.

Linux ext4 uses bitmap + extents + pre-allocation for performance.

8. **Explain the Virtual File System (VFS) architecture and its implementation as a case study in Linux.**

Answer:
Virtual File System (VFS) is an abstraction layer in Linux that allows different concrete file systems to coexist and be accessed uniformly.

Key Objects in Linux VFS:

- Superblock Object: Represents a mounted file system (contains file system type, block size, etc.).
- Inode Object: Represents a file or directory (metadata, permissions, block pointers).
- Dentry (Directory Entry) Object: Represents a name-to-inode mapping (used for dentry cache).
- File Object: Represents an open file (file pointer, open mode).

How VFS Works:

1. User issues system call (e.g., open(), read()).
2. VFS layer receives generic call.
3. VFS looks up the appropriate file system (through registered file_operations, inode_operations, super_operations structures).
4. Calls the specific file system's method.
5. Returns result to user.

Advantages of VFS:

- Uniform interface for all file systems.
- Easy to add new file systems (just register with VFS).
- Supports network file systems (NFS), virtual file systems (proc, sysfs, tmpfs).

Linux Case Study: Linux supports 50+ file systems (ext4, btrfs, xfs, ntfs, vfat, etc.) through VFS. It also maintains dentry cache and inode cache for high performance. SELinux integrates with VFS for mandatory access control.